SCQ Discovery Framework

1. Purpose

SCQ is a structured way to discover the real problem before proposing a technical solution.

- **S - Situation:** What is happening today?
- **C - Complication / Challenge:** Why is the current state insufficient?
- **Q - Question:** What must be answered to design a successful solution?

The older SCQ material focused mainly on cost, time and tool sprawl. The modern framework retains that intent but adds measurable outcomes, stakeholders, constraints, evidence and decision criteria.

2. The rule

Do not start SCQ with a preferred technology.

Bad discovery:

> We should deploy Kubernetes. What Kubernetes features do you need?

Better discovery:

> What workload problem exists today, who experiences it, how often, what is the impact, and what result would make the project successful?

Technology belongs after the problem is understood.

3. Situation

Capture factual current-state information.

Questions

- Who owns the process/system?
- Who uses it?
- What business process does it support?
- What systems/tools are currently involved?
- What is manual vs automated?
- How frequently does the process run?
- What volumes are handled?
- What environments exist?
- What dependencies are external?
- What security or regulatory constraints exist?
- What data is processed and how sensitive is it?
- What evidence exists: logs, tickets, costs, cycle times, outage records, spreadsheets, architecture diagrams?

Output

A short current-state statement based on facts, not assumptions.

Example:

> The analytics team receives a ZIP archive containing multiple Excel workbooks each month. An analyst manually extracts quarterly values from Empirical and Regression sheets, combines them into an output workbook and updates a database. The process depends on spreadsheet interpretation and currently has no automated validation, rerun control or operational monitoring.

4. Challenge / Complication

Describe the measurable impact of the current situation.

Use these dimensions where applicable:

- **Time:** cycle time, waiting time, deployment time, recovery time.

- **Cost:** infrastructure cost, license cost, operational labor, rework.
- **Quality:** errors, inconsistent output, missing validation.
- **Reliability:** outages, manual recovery, fragile dependencies.
- **Security:** exposed credentials, insecure transfer, excessive privileges.
- **Scale:** inability to handle volume, concurrency or growth.
- **Visibility:** lack of metrics, logs, ownership or audit trail.
- **Delivery:** delayed releases, manual handoffs, unrepeatable environments.

A challenge must explain why the project matters.

Example:

> Manual extraction is slow, difficult to audit and vulnerable to incorrect cell/sheet interpretation. Same-day reruns can create duplicate or inconsistent data, and failures are not centrally observable. The business requires a repeatable end-to-end process that completes within 30 minutes and preserves data integrity.

5. Question

Questions must close information gaps that materially affect design or acceptance.

Business questions

- What outcome is mandatory?
- How is success measured?
- What is the acceptable failure rate?
- What is the maximum acceptable processing time?
- Which requirements are Must vs Should?

Data questions

- What is the authoritative source?
- Can source schemas/layouts change?
- How are duplicates identified?
- What is the retention period?
- What is considered sensitive data?
- What is the expected data volume and growth?

Integration questions

- Which APIs, repositories, servers or databases are available?
- What authentication mechanisms are supported?
- What rate limits, quotas or maintenance windows apply?
- What network restrictions exist?

Operational questions

- Who responds to failures?
- What alerts are actionable?
- What recovery time is acceptable?
- How should reruns work?
- What backup/restore capability is required?

Delivery questions

- Which environments are required?
- What is the change-approval process?
- What CI/CD platform is allowed?
- What is the rollback expectation?

6. SCQ output format

Each discovery should end with this concise structure:

Situation

3-8 factual statements describing current state.

Challenge

3-8 statements explaining impact and why change is needed.

Key Questions

A prioritized list of unanswered questions.

Desired Outcome

A measurable statement such as:

> Automate ingestion and transformation of all supported source workbooks, publish validated output and database records, complete a normal run in <=30 minutes, support safe reruns, and provide enough telemetry for an operator to diagnose failures without reading source code.

Constraints

Budget, time, technology, data, network, compliance, availability or team constraints.

Assumptions

Facts temporarily assumed true and requiring validation.

Out of Scope

Explicit exclusions.

Decision Criteria

How alternatives will be compared: security, simplicity, cost, portability, recovery, performance, maintainability, etc.

7. Discovery quality gate

SCQ is complete only when the team can answer:

- What problem are we solving?
- Who benefits?
- What evidence shows the problem exists?
- What measurable outcome defines success?
- What constraints materially shape the solution?
- What is still unknown?
- What is explicitly out of scope?

If these cannot be answered, implementation is premature.

8. Example - tool sprawl

Situation

A team uses several independent tools that produce overlapping operational data. Each tool has separate access control, dashboards and maintenance overhead.

Challenge

Costs are duplicated, incident investigation requires switching between tools, ownership is unclear and data correlation is slow.

Questions

- Which capabilities are genuinely unique?
- Which tools are contractually required?
- What are license and operational costs?
- Which data sources are business critical?
- What retention and compliance requirements exist?
- Can open-source or consolidated alternatives satisfy the same controls?
- What migration risk is acceptable?

Desired outcome

Reduce duplicated operational tooling without reducing required visibility, security or supportability.

9. Template

Use `../templates/SCQ_TEMPLATE.md` for project work.